

Ayush Khaitan

Princeton University

☎ (814) 506 2458 • ✉ ayush.khaitan@princeton.edu
🌐 ayushkhaitanmath.github.io

Research Interests

Automated reasoning and machine learning, Differential Geometry, Combinatorics.

Employment

Princeton University

Peretsman Scully Associate Research Scholar

7/2026-6/2029

Rutgers University, New Brunswick

Hill Assistant Professor

9/2023-6/2026

Education

Penn State University

Ph.D., Mathematics

University Park, PA

2017–2023

Birla Institute of Technology and Science

Bachelor of Engineering

Pilani, India

2011–2015

SJI International

International Baccalaureate, Keppel Magus scholar from India

Singapore

2008–2010

Selected preprints

2025: An LLM-CAS framework for proving asymptotic inequalities with [Vijay Ganesh](#). [arXiv link](#). We solve a question posed by Terence Tao on MathOverflow and his blog. Our [tool](#) was recently [highlighted](#) by Tao.

2025: Elementary symmetric polynomials under the fixed point measure with [Bhargav Narayanan](#) and [Ishan Mata](#) (submitted). [arXiv link](#).

2026: Local and global conformal invariants of submanifolds with [Jeffrey Case](#), [Yueh-Ju Lin](#), [Aaron Tyrell](#) and [Wei Yuan](#) (submitted). [arXiv link](#).

2026: A Lean Formalization of the Hamilton–Perelman Proof of the Three-Dimensional Poincaré Conjecture with [Bennett Chow](#), [Yuan Liao](#) and [Ziyang Qin](#) (submitted). [arXiv link](#)

.

Publications

2022: The weighted ambient metric with [Jeffrey S. Case](#), SIGMA 18 (2022), 086, 21 pages. [Journal Link](#)

2022: GJMS operators of smooth metric measure spaces, *Journal of Geometric Analysis*. [arXiv link](#)

2022: Weighted renormalized volume coefficients, *Differential Geometry and its Applications*. [arXiv link](#)

2023: Ambient metric for manifolds with density and the Ricci flow, *Advances in Mathematics* (2026), Article reference: YAIMA_110787. [Journal link](#).

2024: Computing renormalized curvature integrals on Poincaré-Einstein manifolds with Jeffrey Case, Yueh-Ju Lin, Aaron Tyrell and Wei Yuan, *Advances in Mathematics*. [arXiv link](#).

Research activities

Co-organizer: JMM Panel on the use of AI tools to aid Mathematics research. [Panel website](#). Joint Mathematics Meetings, 2025 in Seattle

Co-organizer: LLMs in Mathematics research group, Rutgers University. [Research group website](#). 2023–Present

Co-organizer: Complex geometry and Harmonic analysis seminar, Rutgers University. [Seminar website](#). 2024–Present

Co-organizer: Special session on recent developments in geometric analysis, AMS Eastern Sectional Meeting 2024, Howard University, Washington D.C. [Meeting website](#). April, 2024

Co-organizer: AI and Math seminar, Rutgers University. 2024–2026. [Seminar website](#).

Co-organizer: AI and Math Lunch seminar, Princeton University, 2026–Present

Teaching

Rutgers

A Lean based Intro to Proofs (Honors course): Rutgers University, two semesters

Machine learning and AI tools in Math Research (Graduate course): Rutgers University, one semester

Linear algebra: Rutgers University, one semester

History of Mathematics: Rutgers University, one semester

Calculus for engineers: Rutgers University, one semester

Penn State

Calculus I: Penn State University, Multiple semesters

Calculus II: Penn State University, Multiple semesters

Linear Algebra: Penn State University, One semester

Plane trigonometry: Penn State University, One semester

Invited Talks

2022: Geometry, Topology and Dynamical Systems seminar, University of Texas at Dallas

2022: Geometry Luncheon seminar, Penn State University

2023: Geometric Analysis Seminar, Rutgers University

2023: Nonlinear Analysis Seminar, Rutgers University

2023: Geometry, Topology and Dynamical Systems seminar, University of Texas at Dallas
2023: Geometry/Topology seminar, Stony Brook University
2024: Colloquium, Rutgers University, Newark
2025: Geometry and Topology seminar, University of Washington, St Louis
2025: Special session on geometric analysis and PDEs, AMS Eastern Sectional Meeting, 2025, Hartford, CT
2025: CUNY Geometric Analysis Seminar, City University of New York, NY
2025: Brown University Geometric Analysis Seminar, Providence, RI
2025: AI and Math seminar, Rutgers University, New Brunswick, NJ
2025: AI Seminar, Simon Fraser University, CA
2025: AI/ML talk, Penn State University, PA
2026: Presentation at Stanford University, CA
2026: Presentation at Kerala School of Mathematics, Kerala, India
2026: Presentation at AMS Spring Eastern Sectional Meeting, Boston College, MA
2026: Stony Brook Differential Geometry Seminar, Stony Brook, NY
2026: Notre Dame Geometric Analysis Seminar, South Bend, IN
2026: Columbia University Geometric Analysis seminar, New York City, NY

Awards

2008: Keppel Magus Scholarship (\$40,000), IB Diploma in Singapore, one scholar from Calcutta, India
2009: Bronze medal, Singapore Math Olympiad
2017: Distinguished Graduate Fellowship, Penn State University, given to 10 incoming graduate students across all departments
2022: Pritchard Dissertation Fellowship, Penn State University, given to 2 graduating PhD students in the Mathematics department
2024-2026: AIM SQuaRE Grant with Jeffrey Case, Yueh-Ju Lin, Aaron Tyrell and Wei Yuan
2025: AMS Travel grant, Mathematical Congress of the Americas
2026: AMS-Simons Travel Grant (2026-2028)